

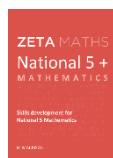
Revision topic list for Form III Mathematics Exam, 30th April 2026

Use the textbook chapters and any questions suggested by your teacher to practise the skills you have learned in Form III.

Zeta textbook

<https://ebooks.zetapress.co.uk/yyrtww>

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Topic	<i>You should be able to...</i>	Chapter	Questions	Revised?
Expanding brackets	Expand two or more single brackets and simplify	5.3		
	Expand double brackets	5.4 – 5.8		
	Expand double brackets – binomial and trinomial	5.11		
Solving equations and inequations	Form linear equations given information	14.7		
	Solve equations with fractions	14.9		
	Solve inequalities	14.8, 14.10		
Right-angled trigonometry	Calculate a side in a right-angled triangle given another side and an angle, including calculating the hypotenuse	Zeta N4 Ch21		
	Calculate an angle in a right-angled triangle given two sides			
	Solve problems involving the use of right-angled trigonometry			
Indices I	Use the first three laws of indices to simplify expressions	3.5 – 3.8		
Standard Form	Carry out calculations with numbers in standard form, <i>without a calculator</i>	4.2		
	Carry out calculations with numbers in standard form <i>using a calculator</i>	4.3		
Factorising I	Factorise expressions using the highest common factor	6.1 – 6.3		
	Identify a difference of two squares and factorise these expressions	6.4		
	Factorise using both the HCF and a difference of two squares	6.5		
	Factorise trinomial expressions (unitary coefficient of x^2)	6.6 – 6.9		
	Identify which technique to use to factorise expressions	6.11		
Quadratics I	Solve quadratic equations by factorising	17.4, 17.5A – C		
Percentages	Increase or decrease quantities by percentages	24.4a		
	Use the terms appreciation and depreciation and calculate the compound interest over a given number of years	24.4b		
	Calculate the original amount given the new amount (reverse percentages)	24.5		

Fractions	Add and subtract mixed number fractions	25.3 – 25.5		
	Multiply and divide mixed number fractions	25.6 – 25.9		
Statistics	Find the quartiles and calculate the interquartile range of a data set	26.2		
Straight Line	Calculate the gradient of the line between two points using the gradient formula	9.2		
	Use the fact parallel lines have the same gradient to solve problems	9.3A		
	Sketch a line given its equation	12.2		
	Find the equation of a straight line given two points, $y = mx + c$ or $y - b = m(x - a)$	12.3		
	Find the gradient and y-intercept of a line by rearranging the equation of the lines	12.4		
	Find the equation of a straight line parallel to another	12.5		
Changing the Subject of Formula	Change the subject of formula involving powers or roots	16.4		
	Change the subject of formula involving multiple operations	16.5 – 16.8		
Rounding	Round numbers to 1, 2 or 3 decimal places	1.1		
	Round numbers to 1, 2 or 3 significant figures	1.2		
Circles – arcs and sectors	Calculate the length of an arc of a circle	10.4		
	Calculate the area of a sector of a circle	10.5		
	Use algebraic skills to calculate the angle, radius or diameter given the arc length or sector area	10.6		
Properties of Shapes	Calculate the interior or exterior angle in polygons	18.1		
	Use the three circle angle properties to find unknown angles in circles	18.2		
Pythagoras' theorem	Use the Converse of Pythagoras' theorem to determine whether a triangle is right-angled	19.2		
	Use Pythagoras' theorem to solve problem involving circles	19.3		
	Use Pythagoras' theorem to solve problems involving 3D shapes	19.5		
Volume of 3D shapes	Calculate the volume of prisms and cylinders	11.1 – 11.2A		
	Calculate the volume of part cylinders	11.2B		
	Volume of pyramids, cones, spheres and hemispheres	11.3 – 11.5		
	Use algebraic skills to calculate height or radius of a 3D shape given its volume	11.5		
	Calculate the volume of composite 3D shapes	11.7		